GENERATIVE AI BIT BANK I MID

1. Which company developed GPT models?
a) Apple **b) OpenAI** c) Google d) Facebook

2. Which technique is often used in generative art creation?
a) Regression **b) GANs** c) Linear models d) SVM

3. What are Variational Auto-encoders used for?
a) Prediction b) Regression **c) Content Generation** d) Data classification

4. Which loss function is key in GANs?
a) Mean squared error **b) Binary cross-entropy** c) Hinge loss d) Euclidean loss

5. What are the two parts of a GAN?
a) Classifier and Clusterer **b) Generator and Discriminator**
c) Predictor and Analyzer d) Encoder and Decoder

6. What does a Discriminator do in GANs?
a) Creates new data b) Deletes old data
c) Evaluates generated data d) Trains the generator

7. A key advantage of transformers over RNNs is:
a) Better for video editing **b) Faster training due to parallelism**
c) Need more data d) Smaller model size

8. Stable Diffusion is used for:

a) Language translation b) Image classification c**) Image generation** d) Noise reduction

9. What is "zero-shot learning" in LLMs?
a) Training with zero data b) Training without GPUs
c) **Answering without task-specific examples** d) Compressing tokens

10. Your RAG system retrieves irrelevant document chunks. What should you check first?
A. The font size of the document **B. Chunking strategy and embedding quality**
C. The model's logo D. The user's browser

11. What is the primary function of Convolutional Neural Networks (CNNs)?
a) Sequence prediction **b) Image classification**
c) Text generation d) Reinforcement learning

12. How many layers are typically found in the architecture of a Feedforward Neural Network (FNN)?
a) One b) Two **c) Three** d) Variable

13. What is the main purpose of pooling in a Convolutional Neural Network?
A To increase the number of trainable parameters
B **To reduce dimensionality while keeping important features**
C To convert images into grayscale
D To normalize pixel values

14. Which pooling method is commonly used in CNNs to reduce the size of feature maps?
A Min pooling B Max pooling C Average pooling D Mean pooling

15. What is the primary function of the "Softmax" function in a Neural Network?
A. To increase the speed of training
B. **To convert outputs into probabilities that sum to 1**
C. To delete unnecessary neurons
D. To store the weights

16. What problem does RAG primarily aim to solve?
A The inability of language models to generate coherent text
B **The outdated knowledge in pre-trained language models**
C The slow processing speed of large language models
D The limited creativity of generative AI systems

17. Which embedding technique is commonly used in RAG systems?
A One-hot encoding **B Dense vector embeddings**
C Binary encodings D Random projections

18. What is a common challenge in RAG systems related to document chunking?
A Ensuring chunks are exactly the same size
B Finding the optimal chunk size to balance context preservation and specificity
C Eliminating all overlap between chunks
D Converting all chunks to the same file format

19. What is "hallucination" in the context of RAG systems?
A When the retrieval system returns irrelevant documents
B **When the language model generates content not supported by retrieved information**
C When the vector database crashes unexpectedly
D When the embeddings fail to capture semantic meaning

20. What is "retrieval fusion" in RAG systems?
    A Merging multiple knowledge bases into one
    B Combining retrieved documents to create a single document
    C Integrating results from multiple retrieval methods
    D Fusing the retriever and generator into a single model

21. The “reward function” in RL is used for:
    a) Defining the neural network architecture
    b) Calculating the probability of actions
    c) Evaluating the quality of an agent’s actions
    d) Filtering noisy observations

22. What is the vanishing gradient problem in deep learning?
    a) A phenomenon where gradients become too large during training
    b) A problem caused by a small learning rate
    c) A situation where gradients approach zero, leading to slow or stalled learning
    d) A challenge in parallelizing deep learning algorithms

23. What is the purpose of the pooling layer in a CNN?
    a) It reduces the spatial dimensions of the feature maps.
    b) It adds more convolutional layers to the network.
    c) It introduces non-linearity to the network.
    d) It performs feature extraction on the input data.

24. Which of these is a transformer-based model?
    A. ResNet
    B. Word2Vec
    C. BERT
    D. VGG

25. Suppose a convolutional neural network is trained on ImageNet dataset (Object recognition dataset). This trained model is then given a completely white image as an input.The output probabilities for this input would be equal for all classes. True or False?

    A. True **B. False**

26._____helps to find the best model that represents our data and how well the chosen model will work in future.

MODEL EVALUATION

27. What type of activation function is typically used in the gates of an LSTM?
    A Sigmoid B Tanh C ReLU **D Softmax**

28. What is a key advantage of using LSTM over a simple RNN for tasks involving long sequences?
A LSTM has fewer parameters than an RNN
B LSTM uses more layers than an RNN
C LSTM can better capture long-range dependencies due to its gating mechanism
D LSTM is faster to train than RNN

29. In TensorFlow, how would you define an LSTM layer with 128 units for time series forecasting?

A tf.keras.layers.LSTM(128)
B tf.keras.layers.LSTM(units=128)
C tf.keras.layers.LSTM(output_dim=128)
D tf.keras.layers.LSTM(128, return_sequences=True)

30. What is the primary purpose of the Keras library in TensorFlow?

A Image and video processing B Data analysis and visualization
C High-level API for neural networks D Data management and storage

31. Which API in Keras is best suited for building complex neural network architectures?

A Sequential API B Model API **C Functional API** D Pre-trained API

32. What is the primary purpose of the LSTM (Long Short-Term Memory) layer?

A Image classification **B Handling long-term dependencies in sequential data**
C Speeding up model training D Reducing model complexity

33. Which scenario best exemplifies the use of one-shot prompting?
a.) Providing a detailed list of instructions followed by multiple examples
b.) Giving one example of a complex task and expecting the model to generalize
c.) Using a large dataset to train the model incrementally
d.) Setting up a reinforcement learning environment with numerous iterations

34. What practice would help reduce hallucinations in an LLM giving factual advice?
a.) Using open-ended questions
b.) Providing specific source references in prompts
c.) Allowing the model to guess the missing data
d.) Encouraging speculative responses

35. Which strategy is least effective in reducing hallucinations in language models?
ANS: Using a smaller dataset for training.

36. A regulated enterprise assistant must strictly follow company policies and refuse to answer when information is missing. Which prompt engineering principle best enforce this behavior?
A. Persona anchoring
B. Higher temperature with self-consistency
C. Explicit refusal conditions and constraints
D. Open-ended role definition

37. You receive a JSON response from an API containing a list of products. You want to analyze prices using pandas. What is the best approach?
A. Manually loop and print values
B. Save JSON to Excel before analysis
C. Use a pivot table without a DataFrame
D. **Convert the JSON list directly into a pandas DataFrame**

38. What is an optimizer in TensorFlow?
A) A function that measures the difference between the predicted output and the actual output
B) A function that computes the gradient of a neural network
C) **A function that updates the weights and biases of a neural network**
D) A function that initializes the weights and biases of a neural network

39. What is transfer learning in TensorFlow?
A) A technique for initializing the weights and biases of a neural network
B) A technique for updating the weights and biases of a neural network
C) A technique for measuring the difference between the predicted output and the actual output
D**) A technique for reusing pre-trained neural network models**

40. How do you generate **predictions** for new data?
Model.predict()